Solution Manual

Analysis and Design of Elementary MOS Amplifier Stages

- First Edition -
Chapter 5

Boris Murmann
September 2013

restart : *with*(*Units*[*Standard*]) : *with*(*plots*) : *interface*(*imaginaryunit* = *j*) : **read** "Parameters.mpl" :

5.1

$V_{DD} := 5[\![V]\!] : W := 20 : R := 4[\![k\Omega]\!] :$

(a) We can compute IIN=ID by first computing VOV:

$$I_D = \frac{V_{DD} - V_{T0n} - V_{OV}[\![V]\!]}{R} :$$

$$\frac{1}{2}\mu nC_{ox} \cdot W \cdot V_{OV}^2[\![V^2]\!] = \frac{V_{DD} - V_{T0n} - V_{OV}[\![V]\!]}{R} :$$

$$\frac{1}{2}\mu nC_{ox} \cdot W \cdot V_{OV}^2[\![V^2]\!] + \frac{1}{R}\ V_{OV}[\![V]\!] - \frac{V_{DD} - V_{T0n}}{R} := 0 :$$

$$V_{OV} := -\frac{1.}{2 \cdot \frac{1}{2}\mu nC_{ox} \cdot W \cdot R} + \sqrt{\left(\frac{1.}{2 \cdot \frac{1}{2}\mu nC_{ox} \cdot W \cdot R}\right)^2 + \frac{V_{DD} - V_{T0n}}{\frac{1}{2}\mu nC_{ox} \cdot W \cdot R}} = 1.271\ [\![V]\!]$$

$$I_D := \frac{V_{DD} - V_{T0n} - V_{OV}}{R} = 0.001\ [\![A]\!] \xrightarrow{\text{replace units}} 807.327\ [\![uA]\!]$$

Check:

$$V_{OV} := V_{DD} - I_D \cdot R - V_{T0n} = 1.271\ [\![V]\!]$$

(b) We repeat this calculation for slow and fast parameters:

$$V_{DD} := 4.5[\![V]\!] : V_{T0nslow} := 0.7[\![V]\!] : \mu nC_{oxslow} := 40\frac{[\![uA]\!]}{[\![V^2]\!]} :$$

$$V_{OVslow} := -\frac{1.}{2 \cdot \frac{1}{2}\mu nC_{oxslow} \cdot W \cdot R} + \sqrt{\left(\frac{1.}{2 \cdot \frac{1}{2}\mu nC_{oxslow} \cdot W \cdot R}\right)^2 + \frac{V_{DD} - V_{T0nslow}}{\frac{1}{2}\mu nC_{oxslow} \cdot W \cdot R}} = 1.260\ [\![V]\!]$$

$$I_{Dslow} := \frac{V_{DD} - V_{T0nslow} - V_{OVslow}}{R} = 0.001\ [\![A]\!] \xrightarrow{\text{replace units}} 635.008\ [\![uA]\!]$$

$$V_{DD} := 5.5[\![V]\!] : V_{T0nfast} := 0.3[\![V]\!] : \mu nC_{oxfast} := 60\frac{[\![uA]\!]}{[\![V^2]\!]} :$$

$$V_{OVfast} := -\frac{1.}{2 \cdot \frac{1}{2}\mu nC_{oxfast} \cdot W \cdot R} + \sqrt{\left(\frac{1.}{2 \cdot \frac{1}{2}\mu nC_{oxfast} \cdot W \cdot R}\right)^2 + \frac{V_{DD} - V_{T0nfast}}{\frac{1}{2}\mu nC_{oxfast} \cdot W \cdot R}} = 1.278\ [\![V]\!]$$

$$I_{Dfast} := \frac{V_{DD} - V_{T0nfast} - V_{OVfast}}{R} = 0.001\ [\![A]\!] \xrightarrow{\text{replace units}} 980.426\ [\![uA]\!]$$

(c) The percent deviations are as follows:

$$100 \cdot \frac{I_{Dslow} - I_{\mathrm{D}}}{I_{\mathrm{D}}} = -21.344$$

$$100 \cdot \frac{I_{Dfast} - I_{\mathrm{D}}}{I_{\mathrm{D}}} = 21.441$$

5.2

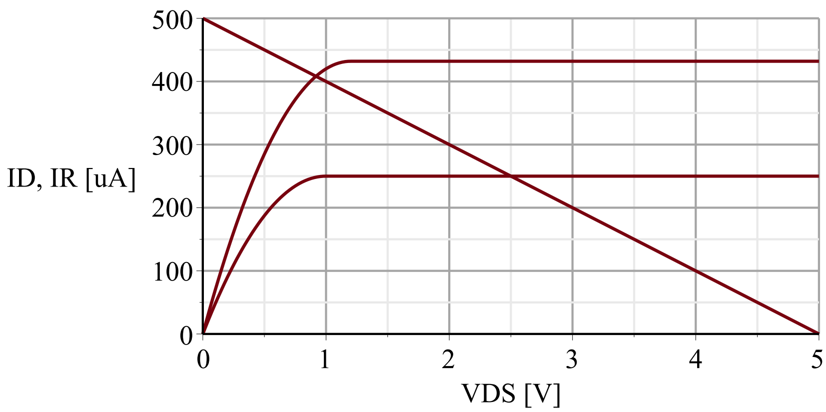
$W := 10 :$

$$IDnom := \textbf{piecewise}\left(VDS > 1.5 - 0.5, \frac{1}{2} \cdot 50 \cdot 10^{-6} \cdot W \cdot (1.5 - 0.5)^2, 50 \cdot 10^{-6} \cdot W \cdot \left(1.5 - 0.5 - \frac{VDS}{2}\right) \cdot VDS\right) :$$

$$IDfast := \textbf{piecewise}\left(VDS > 1.5 - 0.3, \frac{1}{2} \cdot 60 \cdot 10^{-6} \cdot W \cdot (1.5 - 0.3)^2, 60 \cdot 10^{-6} \cdot W \cdot \left(1.5 - 0.3 - \frac{VDS}{2}\right) \cdot VDS\right) :$$

$$IR := VDS \rightarrow \frac{5 - VDS}{10000} :$$

$multiple(plot, [10^6 \cdot \mathrm{IDnom}(VDS), VDS = 0 .. 5], [10^6 \cdot IDfast(VDS), VDS = 0 .. 5], [10^6 \cdot IR(VDS), VDS = 0 .. 5], gridlines, labels = [\text{"VDS [V]"}, \text{"ID, IR [uA]"}])$



For the fast parameters, the load line intersect the output curve in the triode region. The intersect corresponds to the numbers given in Example 5-1.

5.3

Using Eq. (5.25) we find:

$$V_{DS1} = V_{G2}[[V]] - V_{T0n} - V_{OV} = \sqrt{\frac{2\,I_{D6}}{\mu n C_{ox} \cdot \frac{W}{L} \cdot k}} - V_{OV} = V_{OV} \cdot \left(\sqrt{\frac{1}{k}} - 1 \right) :$$

$$r = \frac{V_{DS1}}{V_{OV}} = \sqrt{\frac{1}{k}} - 1 :$$

$$k := Array\left(\left[\frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \frac{1}{8}, \frac{1}{9} \right] \right) :$$

$$\mathrm{r} := \mathrm{k}^{-0.5} - 1 = \left[\begin{array}{cccccc} 1.000 & 1.236 & 1.449 & 1.646 & 1.828 & 2.000 \end{array} \right]$$

5.4

From the discussion of Figure 5-17, we know that the compound circuit behaves like a transistor with a length of L(1+m).

$$V_{DS1} = V_{G2}[\![V]\!] - V_{T0n} - V_{OV} = \sqrt{\frac{2\,I_{D6}}{\mu n C_{ox} \cdot \frac{W}{L \cdot (1+m)}}} - V_{OV} = V_{OV} \cdot \left(\sqrt{1+m} - 1\right) :$$

$$r = \frac{V_{DS1}}{V_{OV}} = \sqrt{1+m} - 1 :$$

$$m := Array([3, 4, 5, 6, 7, 8]) :$$

$$\mathrm{r} := (1+m)^{0.5} - 1 = \left[\begin{array}{cccccc} 1.000 & 1.236 & 1.449 & 1.646 & 1.828 & 2.000 \end{array}\right]$$

5.5

$V_{GS4} = 2\ V_{T0n} + 3\ V_{OV} - \left(V_{T0n} + V_{OV}\right) = V_{T0n} + 2\ V_{OV}$:
$V_{OV4} = 2 \cdot V_{OV}$:

Now, since the gate overdrive voltage scales with sqrt(W/L), the aspect ratio of M4 must be 1/4th that of all other transistors.

5.6

The drain of M5 is at VTn + 2VOV and the source is at VTn + VOV. Hence, VDS5 = VOV.

VIN is given by 2VTn + 3VOV. Hence, VGS5 = VTn + 2VOV, and M5 operates in the triode region with the gate overdrive being twice the drain-source voltage.

From the discussion of Figure 5-17, we know that the W/L of M5 must be 1/3 of that of the other transistors (for the minimum possible output compliance and no margin).

5.7

$R := 'R': W := 'W': m := 'm':$

Note: the wording of this problem was incomplete in the first edition (see errata). The solution below is worked for the revised wording.

We start our analysis using (5.30), which holds in general for the circuit of Figure 5-20 (not just for IIN= IOUT):

$$I_{OUT} \cdot R = V_{OV2} - V_{OV1} :$$

We could now eliminate the VOV terms and solve for IOUT as a function of IIN, but it turns out to be more convenient to work with the gate overdrive voltages at first (to avoid square root heavy equations).

$$m \cdot \frac{1}{2} \mu n C_{ox} \cdot \frac{W}{L} \cdot V_{OV1}^2 \cdot R = V_{OV2} - V_{OV1} :$$

$$m \cdot K \cdot V_{OV1}^2 \cdot R = V_{OV2} - V_{OV1} :$$

$$V_{OV1}^2 + \frac{V_{OV1}}{m \cdot K \cdot R} - \frac{V_{OV2}}{m \cdot K \cdot R} = 0 :$$

The proper solution of this quadratic expression is (we need the plus sign in front of the square root so that VOV1 is positive):

$$V_{OV1} = -\frac{1}{2\, m \cdot K \cdot R} + \sqrt{\left(\frac{1}{2\, m \cdot K \cdot R}\right)^2 + \frac{V_{OV2}}{m \cdot \mathrm{K} \cdot R}} :$$

We can now re-write this result in terms of currents:

$$I_{OUT} = m \cdot K \cdot \left(-\frac{1}{2\, m \cdot K \cdot R} + \sqrt{\left(\frac{1}{2\, m \cdot K \cdot R}\right)^2 + \frac{1}{m \cdot K \cdot R} \cdot \sqrt{\frac{I_{IN}}{K}}}\right)^2 :$$

We now evaluate this expression numerically for the given parameters and show that the intersect with the line IOUT=IIN matches the result given by Eq. (5.33).

$$m := 4 : W := 25 [\![um]\!] : L := 1 [\![um]\!] : R := 2 [\![k\Omega]\!] :$$

$$K := \frac{1.}{2} \mu n C_{ox} \cdot \frac{W}{L} = 0.001 \left[\!\!\left[\frac{A^3\, s^6}{m^4\, kg^2} \right]\!\!\right] \xrightarrow{\text{replace units}} 625.000 \left[\!\!\left[\frac{uA}{V^2} \right]\!\!\right]$$

According to (5.33), we have (for IIN=IOUT=IREF):

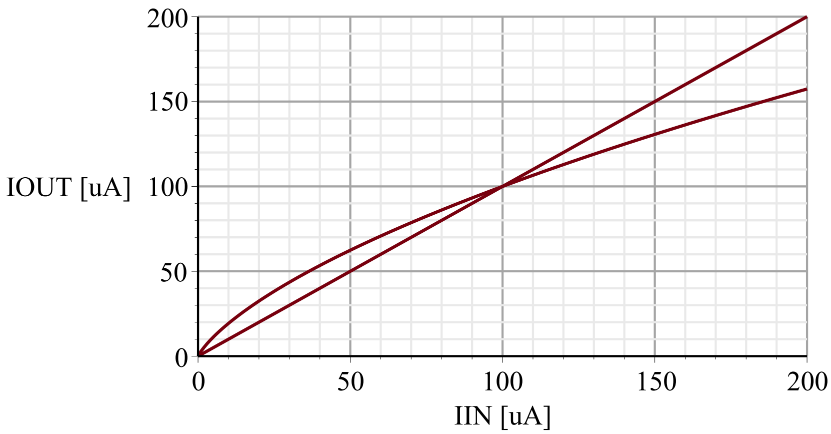
$$I_{REF} := \frac{2 \cdot \left(\sqrt{m} - 1\right)^2}{m} \cdot \frac{1.}{\mu n C_{ox} \cdot \frac{W}{L} \cdot R^2} = 0.000 [\![A]\!] \xrightarrow{\text{replace units}} 100.000 [\![uA]\!]$$

We now generate a plot of IOUT versus IIN to show the intersect occurs as predicted.

$$IOUT := IIN \rightarrow m \cdot K \cdot \left(-\frac{1}{2 \cdot m \cdot K \cdot R} + \sqrt{\left(\frac{1}{2 \cdot m \cdot K \cdot R}\right)^2 + \frac{1}{m \cdot K \cdot R} \cdot \sqrt{\frac{IIN[\![uA]\!]}{K}}} \right)^2 :$$

$$EQUAL := IIN \rightarrow IIN[\![uA]\!] :$$

multiple(*plot*, [IOUT(*IIN*) $\cdot 10^6$, IIN = 0 .. 200], [EQUAL(*IIN*), IIN = 0 .. 200], *gridlines*, *labels* = ["IIN [uA]", "IOUT [uA]"])



5.8

$$V_{T0nfast} := 0.3[\![V]\!] : \mu nC_{oxfast} := 60\frac{[\![uA]\!]}{[\![V^2]\!]} : V_{OUT} := 2.5[\![V]\!] : R_D := 10[\![k\Omega]\!] : V_{DD} := 5[\![V]\!] : W := 10[\![um]\!] : L := 1[\![um]\!] :$$

$$I_D := \frac{V_{DD} - V_{OUT}}{R_D} = 0.000\ [\![A]\!] \xrightarrow{\text{replace units}} 250.000\ [\![uA]\!]$$

$$V_{OV} := \sqrt{\frac{2 \cdot I_D}{\mu nC_{oxfast} \cdot \frac{W}{L}}} = 0.913\ [\![V]\!]$$

$$V_{IN} := V_{T0nfast} + V_{OV} = 1.213\ [\![V]\!]$$

Note that this voltage is quite different from the 1.5V required for nominal conditions. Take-home: never bias the gate of a MOSFET with a constant voltage. The voltage must be "computed" using another MOS device. The voltage gain is now computed as follows:

$$g_{mfast} := \frac{2 \cdot I_D}{V_{OV}} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 0.548\ [\![mS]\!]$$

$$A_{v0fast} := -g_{mfast} \cdot R_D = -5.477$$

Using nominal parameters, the voltage gain is equal to -5. The percent error in the above number is therefore:

$$\frac{A_{v0fast} - (-5)}{-5} \cdot 100 = 9.545$$

5.9

(a) The stated conditions imply the following:

$$V_{OV2} = V_{OV1a} + V_{OV1b} = 2\ V_{OV1}:$$
$$C_{gd1b} + C_{db1b} = C_{gd2} + C_{db2}:$$

Since Cgd and Cdb of a transistor are directly proportional to W, and independent of L, we know that W1=W2. With this in mind, we can now find a relationship between L1 and L2. Since

$$V_{OV} = \sqrt{\frac{2 \cdot I_O}{\mu C_{ox} \frac{W}{L}}}\ :$$

and VOV2 = 2VOV1, we conclude that L2 = 4L1.

(b) We know that

$$R_{o1} = g_{m1} \cdot r_{o1}^2 : \quad \text{(approximately)}$$
$$R_{o2} = r_{o2} :$$

Also, since ro is proportional to L, we know that L2 = 4L1. Therefore:

$$\frac{R_{o1}}{R_{o2}} = \frac{g_{m1} \cdot r_{o1}^2}{4 \cdot r_{o1}} = \frac{g_{m1} \cdot r_{o1}}{4}\ :$$

(c) We are given the following parameters:

$$I_O := 100[\![uA]\!] : W_1 := 10[\![um]\!] : L_1 := 2[\![um]\!] : L_2 := 4 \cdot L_1 :$$

$$V_{OV1} := \sqrt{\frac{2. \cdot I_O}{\mu n C_{ox} \cdot \frac{W_1}{L_1}}} = 0.894\ [\![V]\!]$$
$$V_{Omin} := 2 \cdot V_{OV1} = 1.789\ [\![V]\!]$$
$$g_{m1} := \frac{2 \cdot I_O}{V_{OV1}} = 0.000\ [\![S]\!] \xrightarrow{\text{replace units}} 0.224\ [\![mS]\!]$$
$$\lambda_1 := \frac{\lambda_{Ln}}{L_1} = 0.050 \left[\!\!\left[\frac{1}{V}\right]\!\!\right]$$
$$r_{o1} := \frac{1}{\lambda_1 \cdot I_O} = 2.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 200.000\ [\![kOmega]\!]$$
$$\lambda_2 := \frac{\lambda_{Ln}}{L_2} = 0.013 \left[\!\!\left[\frac{1}{V}\right]\!\!\right]$$

$$r_{o2} := \frac{1}{\lambda_2 \cdot I_O} = 8.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 800.000\ [\![kOmega]\!]$$

$$R_{o1} := g_{m1} \cdot r_{o1}^2 = 8.944\ 10^6\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 8.944\ [\![MOmega]\!]$$

$$R_{o2} := r_{o2} = 8.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 800.000\ [\![kOmega]\!]$$

5.10

$I_{OUT} := 100 [\![uA]\!] : W_1 := 50 [\![um]\!] : L_1 := 1 [\![um]\!] :$

Since M3, M4 and M5 have the same size, we know that the drain currents of M1 and M2 must be equal to IOUT.

$$V_{SG1} := -V_{T0p} + \sqrt{\frac{2. \cdot I_{OUT}}{\mu p C_{ox} \cdot \frac{W_1}{L_1}}} = 0.900 \ [\![V]\!]$$

$$R := \frac{V_{SG1}}{I_{OUT}} = 9000.000 \ [\![\Omega]\!] \xrightarrow{\text{replace units}} 9.000 \ [\![kOmega]\!]$$

5.11

$I_{ref} := 200[\![uA]\!] : W := 50[\![um]\!] : L := 1[\![um]\!] :$

(a) We begin by computing the gate overdrive voltage of M1 (which is also the gate overdrive voltage of all other transistors)

$$V_{OV1} := \sqrt{\frac{2.\cdot I_{ref}}{\mu nC_{ox}\cdot\frac{W}{L}}} = 0.400\ [\![V]\!]$$

$$V_{DS1} := 1.5\cdot V_{OV1} = 0.600\ [\![V]\!]$$

The voltage across the resistor is given by

$$V_R = V_{DS1} + V_{GS2}[\![V]\!] - V_{GS1}[\![V]\!] = V_{DS1} :$$

$$R := \frac{V_{DS1}}{I_{ref}} = 3000.000\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 3.000\ [\![kOmega]\!]$$

(b) A 10mV change is small relative to the gate overdrive of 400mV. We can use the MOSFET's transconductance to estimate the change in current.

$$g_{m1} := \frac{2\cdot I_{ref}}{V_{OV1}} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 1.000\ [\![mS]\!]$$

$$dI := -10[\![mV]\!]\cdot g_{m1} = -0.000\ [\![A]\!] \xrightarrow{\text{replace units}} -10.000\ [\![uA]\!]$$

The corresponding percent error is:

$$\frac{dI}{I_{ref}}\cdot 100 = -5.000$$

(c) In this case the drain voltage of M1 decreases by about 10mV. Again, it is most convenient here to work with the small-signal approximation. One can also work this out using the large signal lambda model.

$$gmro := 50 :$$

$$r_{o1} := \frac{gmro}{g_{m1}} = 50000.000\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 50.000\ [\![kOmega]\!]$$

$$dI := -\frac{10[\![mV]\!]}{r_{o1}} = -2.000\ 10^{-7}\ [\![A]\!] \xrightarrow{\text{replace units}} -0.200\ [\![uA]\!]$$

The corresponding percent error is:

$$\frac{dI}{I_{ref}}\cdot 100 = -0.100$$

Another way to work this problem is to realize that the compound transconductance of M2+M1 is equal to Gm = gm2/(1+gm2*ro1), and multiply the voltage change with this Gm.

5.12

$I_{D1} := 50 [\![uA]\!] : W_1 := 4 [\![um]\!] : W_2 := 4 [\![um]\!] : W_3 := 0.5 [\![um]\!] : L := 1 [\![um]\!] : V_{DD} := 5 [\![V]\!] :$

(a) We begin by computing the gate overdrive voltages:

$$V_{OV1} := \sqrt{\frac{2.\cdot I_{D1}}{\mu nC_{ox}\cdot \frac{W_1}{L}}} = 0.707\ [\![V]\!]$$

$$V_{OV2} := V_{OV1} :$$

$$V_{OV3} := \sqrt{\frac{2.\cdot I_{D1}}{\mu nC_{ox}\cdot \frac{W_3}{L}}} = 2.000\ [\![V]\!]$$

The minimum value for VG2 is:

$$V_{G2min} := V_{OV1} + V_{T0n} + V_{OV2} = 1.914\ [\![V]\!]$$

The maximum value for VG2 can be found by considering the maximum source voltage of M2 that keeps M2 saturated:

$$V_{S2max} := V_{DD} - V_{OV3} - V_{OV2} = 2.293\ [\![V]\!]$$

The maximum VG2 is therefore:

$$V_{G2max} := V_{S2max} + V_{T0n} + V_{OV2} = 3.500\ [\![V]\!]$$

(b) With backgate effect included, the only change is that we can no longer use VT0n, but now need to compute the threshold voltage at the respective source potentials.

$$V_{SB1} := V_{OV1} = 0.707\ [\![V]\!]$$

$$V_{Tn1} := V_{T0n} + \gamma_p\cdot\left(\sqrt{2\cdot\phi_f + V_{SB1}} - \sqrt{2\cdot\phi_f}\right) = 0.700\ [\![V]\!]$$

$$V_{G2min} := V_{OV1} + V_{Tn1} + V_{OV2} = 2.114\ [\![V]\!]$$

$$V_{SB2} := V_{S2max} = 2.293\ [\![V]\!]$$

$$V_{Tn2} := V_{T0n} + \gamma_p\cdot\left(\sqrt{2\cdot\phi_f + V_{SB2}} - \sqrt{2\cdot\phi_f}\right) = 1.019\ [\![V]\!]$$

$$V_{G2max} := V_{S2max} + V_{Tn2} + V_{OV2} = 4.019\ [\![V]\!]$$

5.13

First note that M6 operates in the saturation region and M7 and M8 operate in the triode region. The gate voltage of the transistors is equal to 3*VOV8+VTn. The gate overdrive of M8 is therefore equal to 3* VOV8, and the gate overdrive of M7 is equal to 2*VOV8. Since all drain currents are equal, we can write:

$$\frac{1}{2}\left(\frac{W}{L}\right)_6 V_{OV6}^2 = \left(\frac{W}{L}\right)_7 \cdot \left(2 \cdot V_{OV6} \cdot V_{OV6} - \frac{1}{2} V_{OV6}^2\right) = \left(\frac{W}{L}\right)_8 \cdot \left(3 \cdot V_{OV6} \cdot V_{OV6} - \frac{1}{2} V_{OV6}^2\right) :$$

$$\mathrm{W}_6 V_{OV6}^2 = \mathrm{W}_7 \cdot 3 V_{OV6}^2 = \mathrm{W}_8 \cdot 5 V_{OV6}^2 :$$

$$\frac{\mathrm{W}_7}{W_6} = \frac{1}{3} :$$

$$\frac{\mathrm{W}_8}{W_6} = \frac{1}{5} :$$

5.14

$I_{OUT} := 100[\![uA]\!]$: $V_{OV} := 300[\![mV]\!]$: $L := 3[\![um]\!]$:

The parameter m can be chosen freely in this problem. Let us use $m := 4$:

$$R := \frac{V_{OV} \cdot \left(1 - \frac{1.}{\sqrt{m}}\right)}{I_{OUT}} = 1500.000 \ [\![\Omega]\!] \xrightarrow{\text{replace units}} 1.500 \ [\![kOmega]\!]$$

$$W_2 := \frac{I_{OUT}}{\frac{1.}{2} \cdot \mu nC_{ox} \cdot \frac{1}{L} \cdot V_{OV}^2} = 0.000 \ [\![m]\!] \xrightarrow{\text{replace units}} 133.333 \ [\![um]\!]$$

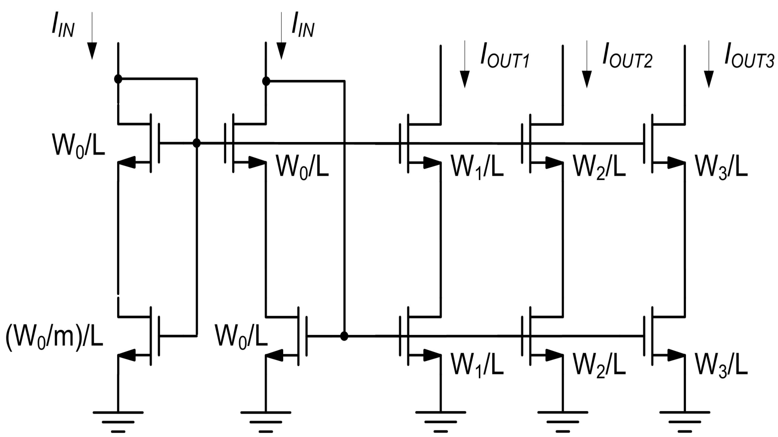
$$W_3 := \frac{I_{OUT}}{\frac{1.}{2} \cdot \mu pC_{ox} \cdot \frac{1}{L} \cdot V_{OV}^2} = 0.000 \ [\![m]\!] \xrightarrow{\text{replace units}} 266.667 \ [\![um]\!]$$

$$W_1 := m \cdot W_2 = 0.001 \ [\![m]\!] \xrightarrow{\text{replace units}} 533.333 \ [\![um]\!]$$

5.15

$I_{IN} := 10[\![uA]\!] : I_{OUT1} := 20[\![uA]\!] : I_{OUT2} := 50[\![uA]\!] : I_{OUT3} := 100[\![uA]\!] : V_{OUTmin} := 0.8[\![V]\!] :$

There are many possible solutions. The one shown below is based on the circuit of Figure 5-18.



We begin by choosing $L := 2[\![um]\!] :$ for reasonable output resistance (minimum length devices are typically avoided in current mirrors).

For reasonable saturation margin of the bottom transistors, we choose $m := 5 :$ From Table 5-6, we know that the VDS of the bottom devices equals 1.45*VOV. To achieve the given compliance voltage with the maximum possible VOV, we have:

$$V_{OV} := \frac{V_{OUTmin}}{1 + 1.45} = 0.327\ [\![V]\!]$$

We can now compute all device widths:

$$W_0 := \frac{I_{IN}}{\frac{1.}{2} \cdot \mu n C_{ox} \cdot \frac{1}{L} \cdot V_{OV}^2} = 0.000\ [\![m]\!] \xrightarrow{\text{replace units}} 11.255\ [\![um]\!]$$

We round up to the nearest full micron (to snap onto a reasonable layout grid):

$W_0 := 12[\![um]\!] :$
$W_1 := 2 \cdot W_0 = 24\ [\![um]\!]$
$W_2 := 5 \cdot W_0 = 60\ [\![um]\!]$
$W_3 := 10 \cdot W_0 = 120\ [\![um]\!]$

Note that these widths are usually realized by connecting the respective number of 12 um unit devices in parallel.

5.16

Shown below is one possible option. Note that the unit devices were split into two so that even the smallest unit has two fingers. This allows sharing of all the drain contacts for all devices in a compact arrangement. In advanced technologies, it is advisable to place "dummy" transistors at the ends of such arrays to avoid "edge effects" (beyond the scope of this module).